DUETGig

A Student Freelancing and Peer Service Platform

Introduction

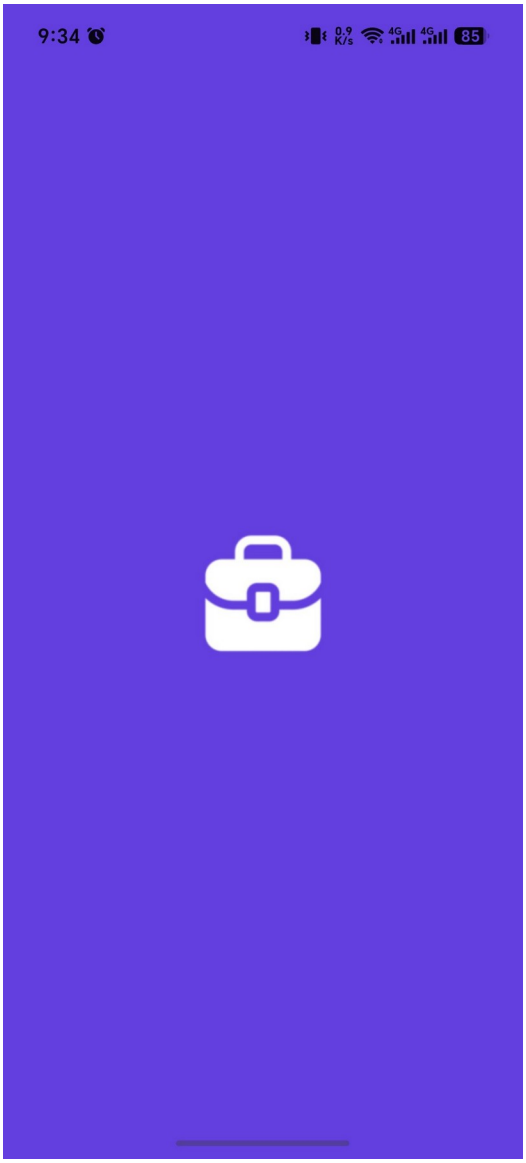
DUETGig is an Android-based mobile platform designed exclusively for the students of Dhaka University of Engineering & Technology (DUET). It bridges the gap between students who possess valuable technical skills (programming, app development, CAD design, robotics, and more) and those who require affordable academic and technical services. The application is developed using Android Studio and Java. Firebase (Authentication, Cloud Firestore, Storage and Cloud Messaging) is used for cloud-based data storage, authentication and real-time notifications. The platform supports the complete journey of a student freelancer: Skill → Experience → Portfolio → Income.

Objective:

- Create an Exclusive Ecosystem for DUET students
- Enable Financial Empowerment through technical and academic skills
- Support Skill & Portfolio Building before graduation
- Encourage Collaboration and knowledge sharing among peers
- Provide a Secure Marketplace for hiring and freelancing
- Enhance University Prestige by showcasing student expertise

---

Splash Screen

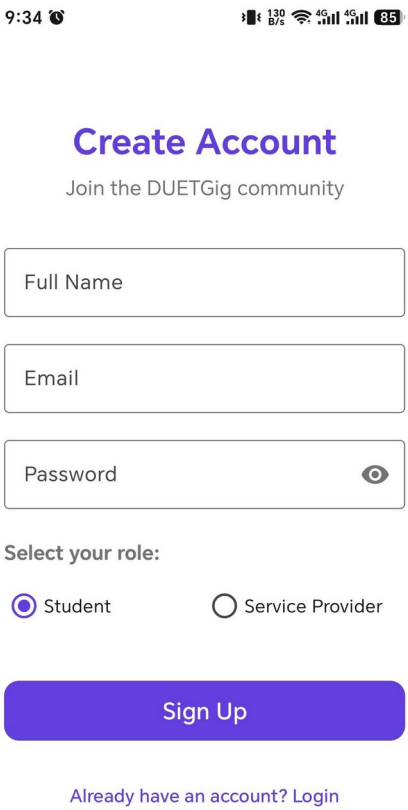


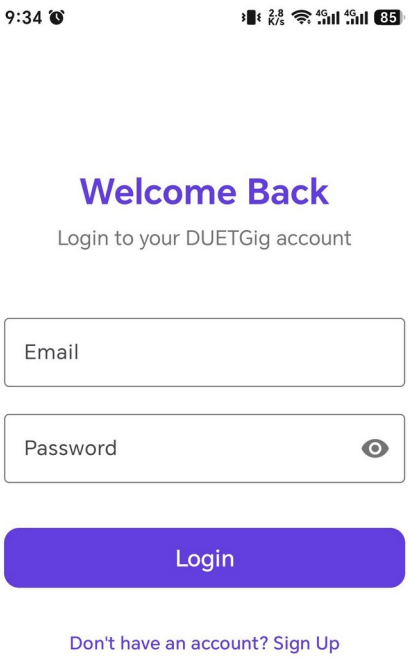
Splash Screen

The application opens with a simple branded splash screen displaying the DUETGig briefcase logo on a solid purple background. It gives the app a professional identity while Firebase initializes the session in the background.

- Branded loading screen
- Automatic session/session-state check
- Smooth transition to Login or Home

Account Creation & Login:



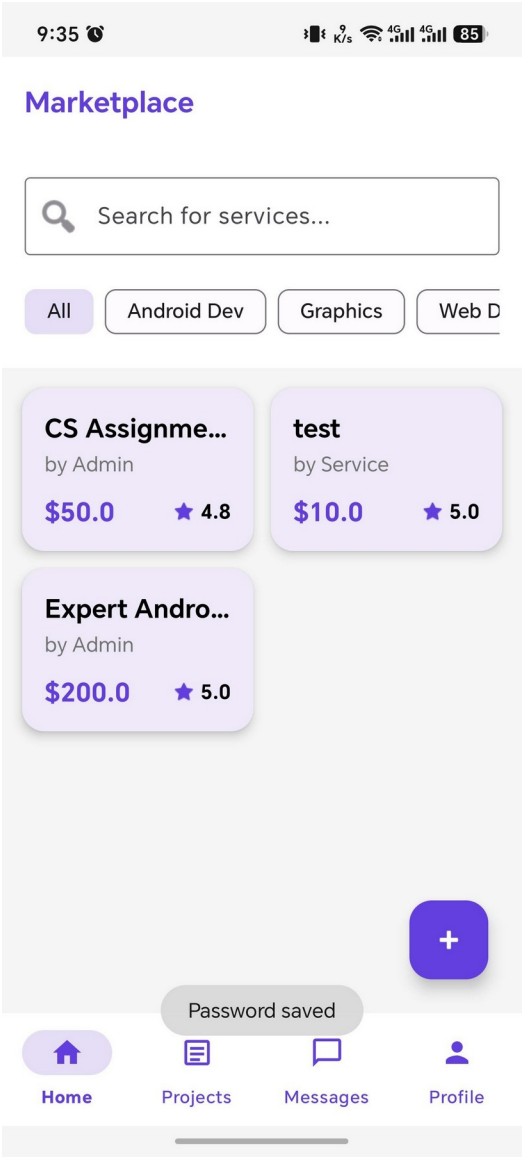


Create Account | **Welcome Back (Login)**

New users can create an account with their full name, email and password, and choose their role — Student or Service Provider. Returning users log in securely with their registered email and password through Firebase Authentication.

- Sign up with name, email and password
- Select role: Student or Service Provider
- Secure login with Firebase Authentication
- Password visibility toggle

Marketplace:

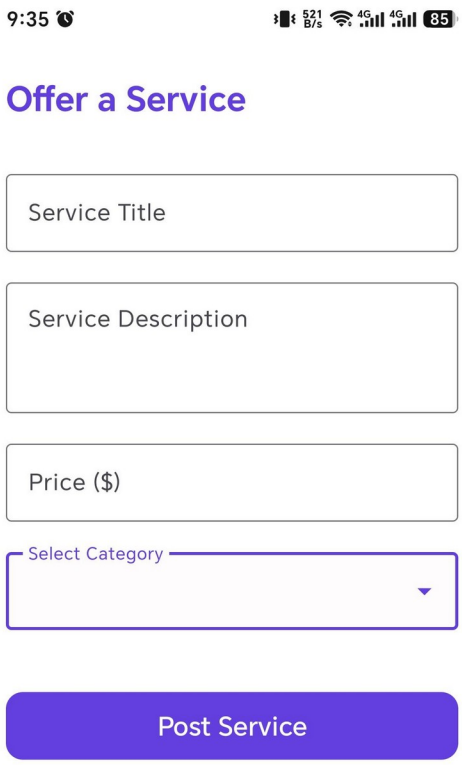


Marketplace / Home

The Home screen shows the Marketplace where all posted services are listed as cards with price and rating. Users can search for services and filter them by category (Android Dev, Graphics, Web Dev, and more).

- Browse all available services
- Search for services by keyword
- Filter by category
- View price and rating at a glance

Offer a Service:

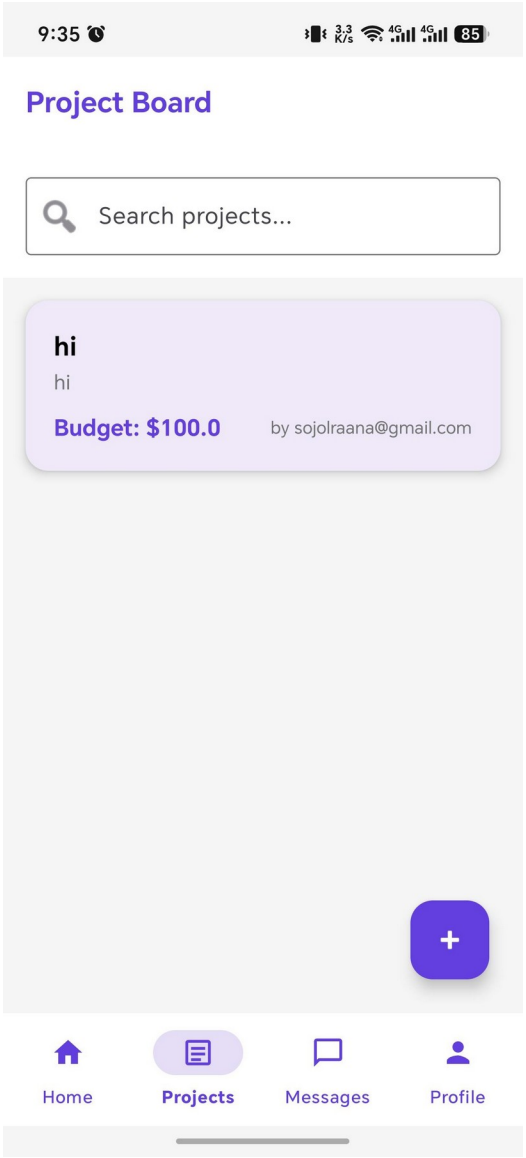


Offer a Service

Service providers can list their own skill-based service by entering a title, description, price and category. Once posted, the service becomes visible to every student in the Marketplace.

- Add service title and description
- Set a price for the service
- Select a category
- Post instantly to the Marketplace

Project Board:

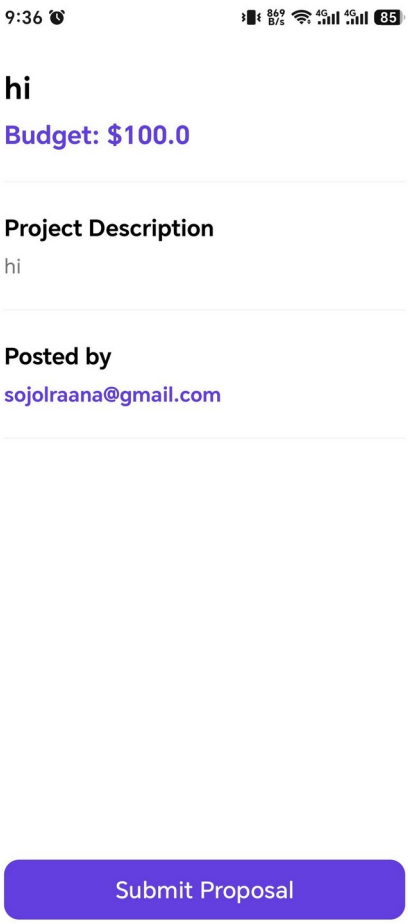


Project Board

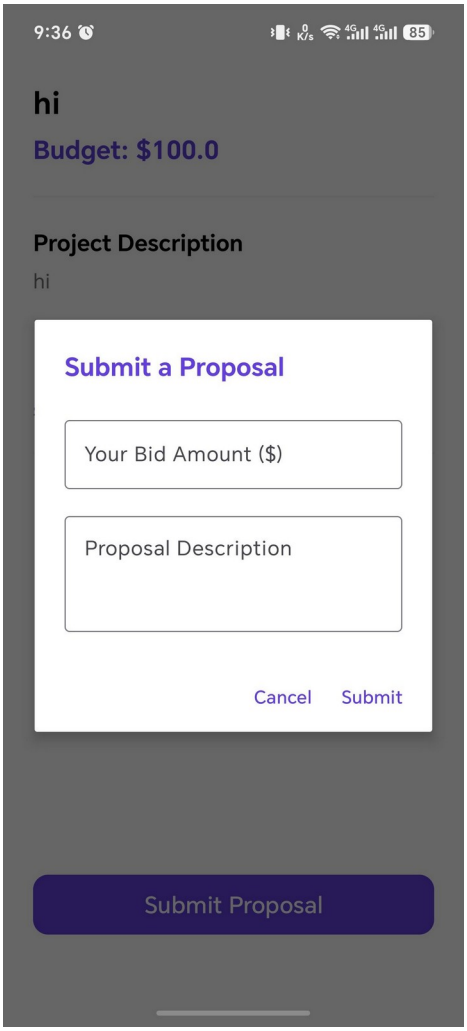
Students who need help can post a custom project with a title, description and budget on the Project Board. Other students can search existing projects and view who posted them before submitting a proposal.

- Post a custom project request
- Search projects by keyword
- View budget and posted-by information
- Quick-add button to create a new project

Project Details & Submit Proposal:



Project Details

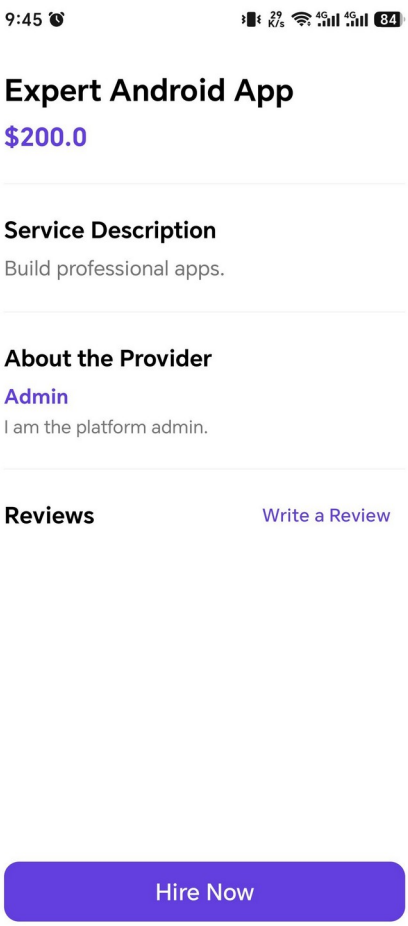


Submit a Proposal

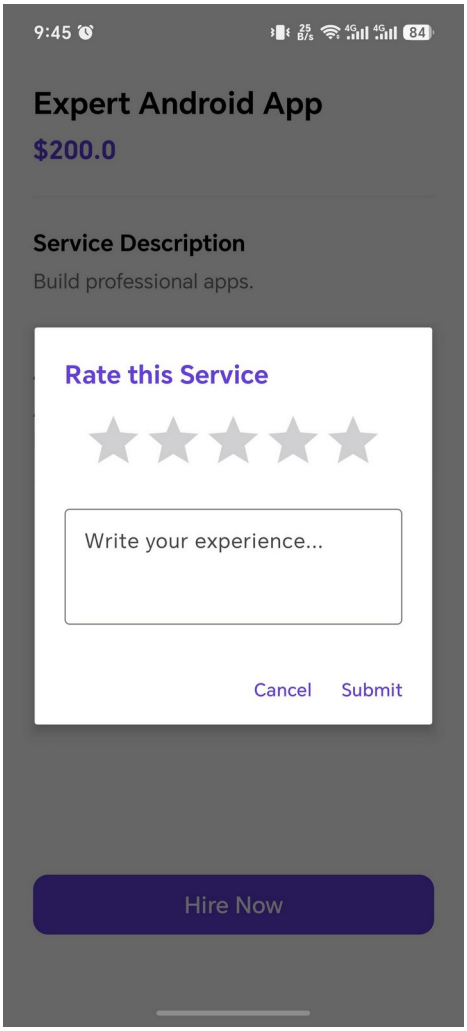
Opening a project shows its full description, budget and the poster's contact email. Interested students tap Submit Proposal to open a form where they enter their bid amount and a short proposal description before sending it to the project owner.

- View full project description and budget
- See who posted the project
- Submit a bid amount
- Write a short proposal description

Service Details, Ratings & Reviews:



Service Details & Reviews

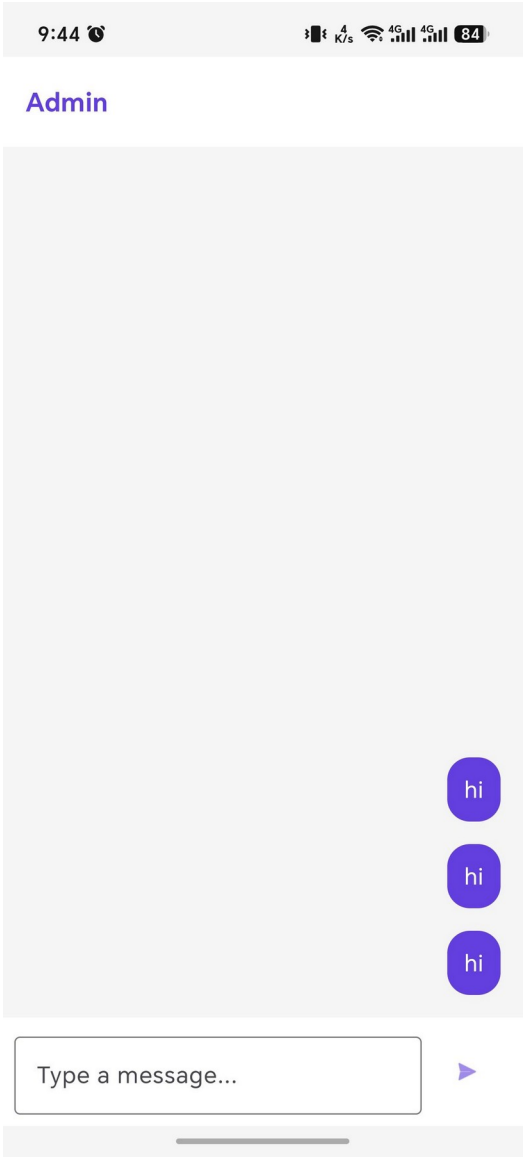


Rate this Service

Each service page shows its description, price, a short bio of the provider, and a Hire Now button. Students who have used a service can leave a star rating and a written review, which helps build trust within the DUET community.

- View service description and provider bio
- Hire the service provider instantly
- Rate a service from 1 to 5 stars
- Write a text review of the experience

Real-Time Chat System:

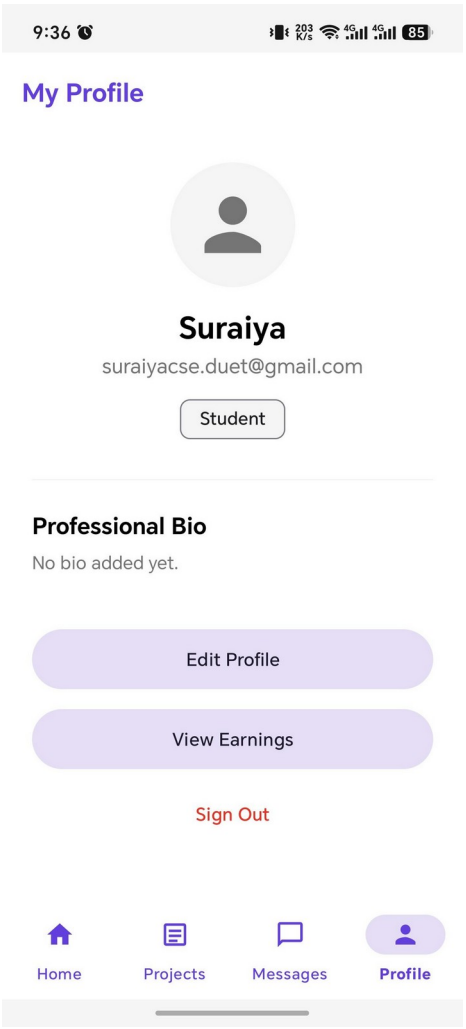


Chat with Provider/Admin

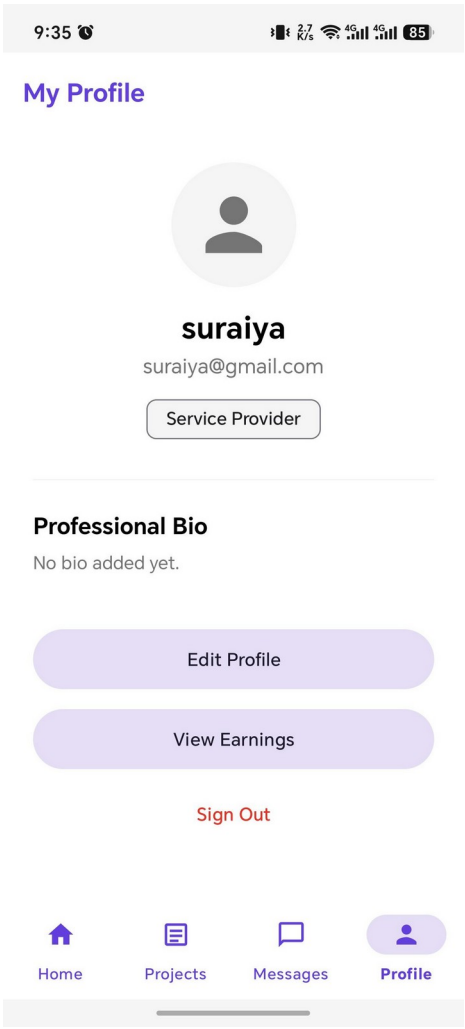
DUETGig includes an integrated real-time chat so clients and freelancers can communicate directly inside the app once a project or service conversation begins, without sharing personal contact details.

- One-to-one real-time messaging
- Message bubbles with timestamps
- Built on Firebase for instant delivery
- Keeps all project communication in one place

Profile Management:



Student Profile

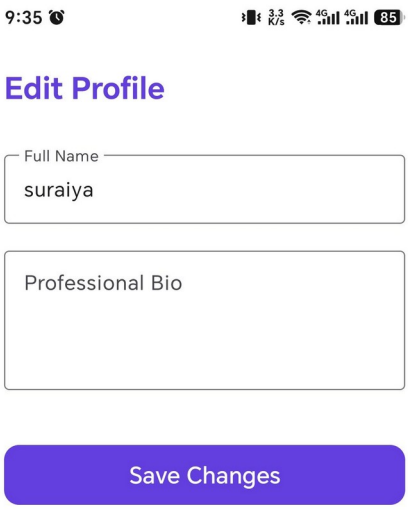


Service Provider Profile

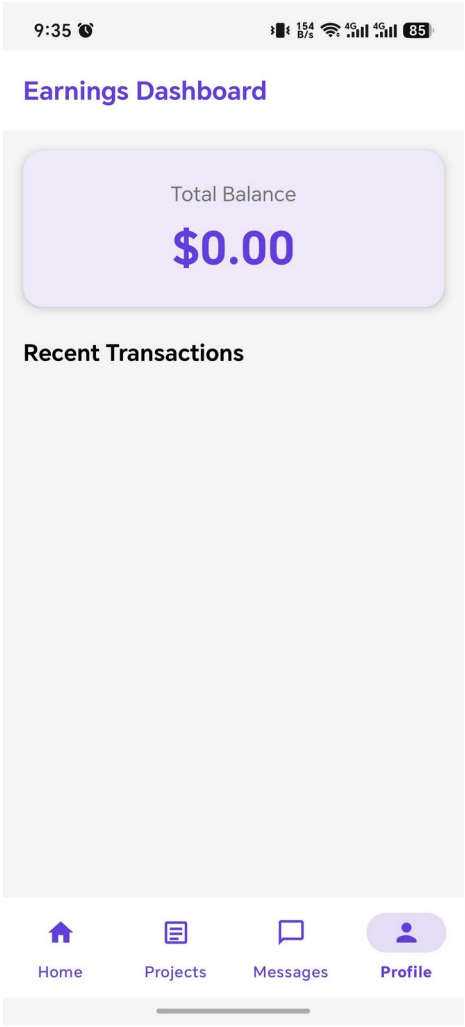
Every user has a profile page showing their name, email and role — Student or Service Provider — along with a professional bio. The profile screen also links to editing details and viewing earnings.

- Display name, email and role badge
- Professional bio section
- Quick links to Edit Profile and Earnings
- Sign out option

Edit Profile & Earnings Dashboard:



Edit Profile

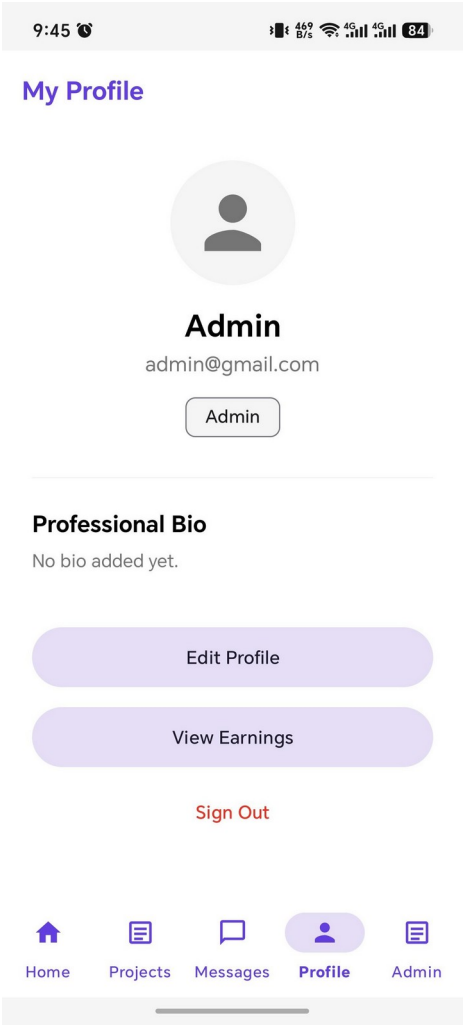


Earnings Dashboard

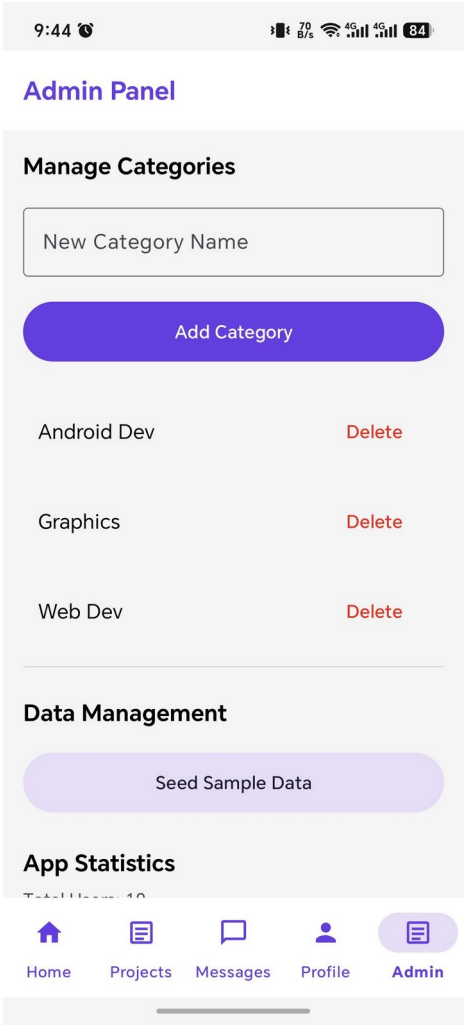
Users can update their full name and professional bio from the Edit Profile screen. Service providers can track their income through the Earnings Dashboard, which shows total balance and a list of recent transactions.

- Edit full name and bio
- Save changes instantly to Firestore
- View total balance
- Track recent transactions

Admin Panel:



Admin Profile

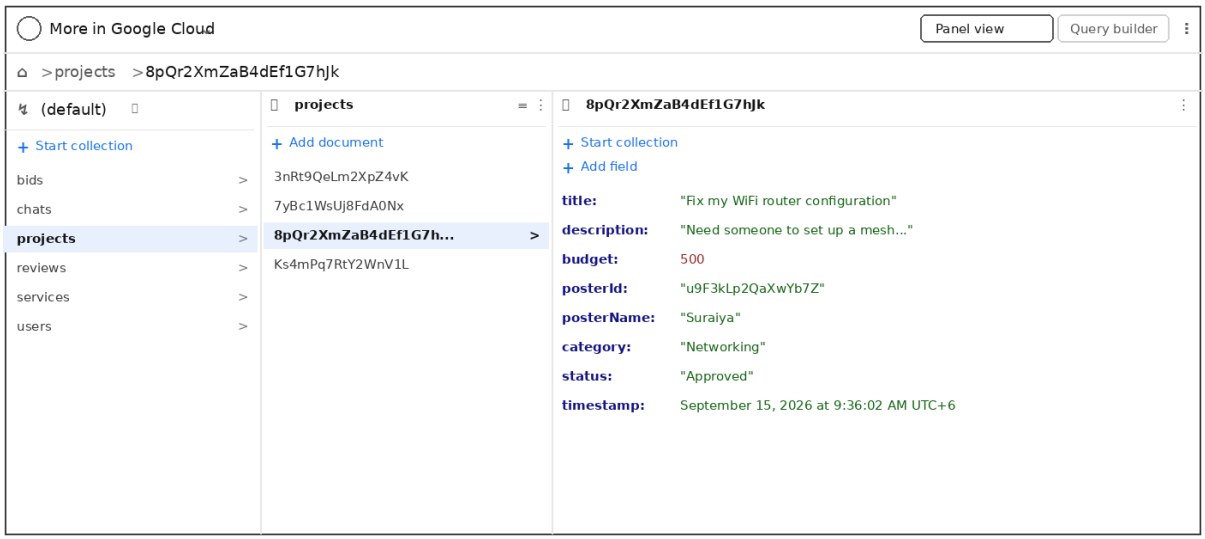


Manage Categories

An Admin role unlocks an extra Admin tab for platform moderation. Admins can add or delete marketplace categories, seed sample data for testing, and view overall app statistics such as total registered users.

- Dedicated Admin tab and role badge
- Add or delete service categories
- Seed sample data for testing
- View app-wide statistics

Firestore Database Record:



Firebase Console — projects Collection

The Firebase Console shows the projects collection and a stored project document. The displayed record contains title, description, budget, posterId, posterName, category, status and timestamp fields. This confirms that the application successfully sends project-related data to the Firestore database.

Database Structure (Firebase Firestore):

DUETGig uses Cloud Firestore as its NoSQL cloud database. Data is organized into the following top-level collections, each storing documents that map directly to the app's core models.

Collection: users

Stores every registered account and their role/approval state.

Field	Type	Description
uid	String	Firebase Auth user ID (document ID)
name	String	Full name
email	String	Registered email address
role	String	"Client", "Service Provider" or "Admin"
bio	String	Professional bio
status	String	"Pending", "Approved" or "Rejected"
isAdmin	Boolean	Whether the user has admin privileges

Collection: projects

Stores project requests posted on the Project Board.

Field	Type	Description
projectId	String	Unique project ID (document ID)
title	String	Project title
description	String	Project description

Field	Type	Description
budget	Number	Budget in USD
posterId / posterName	String	ID and name of the student who posted it
category	String	Project category
status	String	"Pending", "Approved" or "Rejected"
timestamp	Timestamp	Date/time the project was posted

Collection: services

Stores services offered by service providers in the Marketplace.

Field	Type	Description
serviceId	String	Unique service ID (document ID)
title / description	String	Service title and description
price	Number	Price in USD
category	String	Service category
providerId / providerName / providerBio	String	Provider details
rating	Number	Average star rating
status	String	"Pending", "Approved" or "Rejected"

Collection: bids

Stores proposals submitted by students against a posted project.

Field	Type	Description
bidId	String	Unique bid ID (document ID)
projectId	String	Linked project
bidderId / bidderName	String	Who submitted the bid
amount	Number	Proposed bid amount
proposal	String	Proposal description
status	String	"Pending", "Accepted" or "Rejected"
timestamp	Timestamp	Date/time submitted

Collection: chats / messages

Stores one-to-one conversations and their messages for the real-time chat system.

Field	Type	Description
chatId	String	Unique chat ID (document ID)
userIds	List<String>	The two participants in the chat
userNames	Map	userId → display name lookup
lastMessage / lastTimestamp	String / Timestamp	Preview shown in the chat list
messages/{messageId}	Subcollection	senderId, content, timestamp per message

Collection: reviews

Stores star ratings and written feedback left on a service.

Field	Type	Description
reviewId	String	Unique review ID (document ID)
serviceId	String	Linked service
reviewerName	String	Name of the reviewer
rating	Number	Star rating (1–5)
comment	String	Written review text
timestamp	Timestamp	Date/time submitted

Future Scope

- Integrated payment gateways
- AI-based skill recommendations & matchmaking
- Internship matching and employer verification
- Alumni mentorship programs
- Multilingual support and web platform integration
- Certificate issuance for completed milestones

Conclusion

DUETGig provides a centralized, secure platform exclusively for DUET students to turn their technical skills into real income and experience. From posting services and projects to real-time chat, ratings, earnings tracking and admin moderation, the app covers the complete freelancing workflow. Built with Java, Android Studio and Firebase, DUETGig demonstrates an effective mobile-based solution for student economic empowerment, ultimately contributing to a stronger, more connected DUET community.